

PanditConnect

Book Trusted Pandit Ji — Online

Product Requirements & Technical Specification

A multi-sided marketplace for poojas, sanskars & religious ceremonies

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Document Control

Revision History

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0.1	May 2026	Founder	Initial concept note and feature brainstorm
1.0	Jun 2026	Product & Eng	Formalised PRD: architecture, data model, edge-case catalogue, review subsystem, MVP scope

Approvals

Role	Name	Decision	Date
Product Owner	—	Pending	—
Engineering Lead	—	Pending	—
Design Lead	—	Pending	—

How to Read This Document

This specification is the single source of truth for the PanditConnect MVP. Sections 1–8 give product and architectural context; Sections 9–12 define behaviour and data; Section 13 catalogues edge cases and expected handling; Sections 14–20 cover non-functional requirements, scope and planning. Each functional feature lists acceptance criteria that QA can test against. Open questions are tracked in Section 19.

Table of Contents

Document Control.....	1
Revision History.....	1
Approvals.....	1
How to Read This Document.....	1
1. Executive Summary.....	5
2. Product Vision.....	6
2.1 Vision Statement.....	6
2.2 Mission.....	6
2.3 Product Principles.....	6
3. Problem Statement & Market Context.....	7
3.1 Current Problems.....	7
3.2 Target Market.....	7
3.3 Why Now.....	7
4. Solution Overview.....	8
The end-to-end loop.....	8
5. Goals & Success Metrics.....	9
5.1 Business Goals.....	9
5.2 Key Performance Indicators (KPIs).....	9
5.3 North Star Metric.....	9
6. Target Users & Personas.....	10
6.1 Persona 1 — Customer (Devotee).....	10
6.2 Persona 2 — Pandit Ji (Service Provider).....	10
6.3 Persona 3 — Admin (Platform Operator).....	10
6.4 Supported Pooja & Sanskar Catalogue (launch set).....	10
7. System Architecture.....	12
7.1 Architecture Diagram.....	12
7.2 Component Responsibilities.....	12
7.3 Technology Stack & Rationale.....	13
8. User Roles & Access Control (RBAC).....	14
9. Feature Specifications.....	15
9.1 Authentication & Onboarding.....	15
Requirements.....	15
Acceptance Criteria.....	15
9.2 Pandit Registration & Verification.....	15
Required fields.....	15
Verification workflow.....	15
Acceptance Criteria.....	16
9.3 Customer Registration.....	16
9.4 Search & Discovery.....	16
Behaviour.....	16
Profile card contents.....	16
9.5 Ranking Algorithm.....	16
Acceptance Criteria.....	17
9.6 Pandit Public Profile.....	17
9.7 Booking System.....	17
Booking flow.....	17
Booking state machine.....	17
Acceptance Criteria.....	17

9.8 Pandit Dashboard.....	18
9.9 Notifications.....	18
9.10 Internationalisation (i18n).....	18
10. Customer Review & Rating System.....	19
10.1 Objectives.....	19
10.2 Rating Dimensions.....	19
10.3 Eligibility & Rules.....	19
10.4 Review Lifecycle.....	20
10.5 Rating Aggregation & Anti-Gaming.....	20
10.6 Pandit Response.....	20
10.7 Moderation & Abuse Prevention.....	21
10.8 Display & UX.....	21
10.9 Sample Reviews (illustrative).....	21
11. User Journey Flows.....	23
11.1 Customer Journey.....	23
11.2 Customer Flow (detailed).....	23
11.3 Pandit Onboarding Flow.....	23
12. Data Model.....	24
12.1 Entities & Relationships.....	24
12.2 Collection Schemas.....	24
users.....	24
pandits.....	24
poojas.....	25
materials.....	25
bookings.....	25
reviews.....	26
payments (Phase 2).....	26
12.3 Indexing & Integrity Strategy.....	26
13. Edge Cases & Error Handling.....	27
13.1 Authentication & Accounts.....	27
13.2 Registration & Verification.....	27
13.3 Search & Ranking.....	28
13.4 Booking.....	28
13.5 Reviews.....	29
13.6 Dashboard & Revenue.....	29
13.7 Data Integrity & Concurrency.....	30
13.8 Media & Uploads.....	30
13.9 Internationalisation.....	30
13.10 Infrastructure & Free-Tier Limits.....	30
14. Security, Privacy & Compliance.....	32
14.1 Data Protection.....	32
14.2 Identity (Aadhaar) & PII Handling.....	32
14.3 Application Security.....	32
14.4 Content & Trust Safety.....	32
15. Non-Functional Requirements.....	33
16. MVP Scope.....	34
16.1 Must Have.....	34
16.2 Nice to Have (later).....	34
16.3 Explicitly Out of Scope for the MVP.....	34
17. Monetisation & Roadmap.....	35
17.1 Revenue Model (phased).....	35

17.2 Phase-2 Feature Candidates.....	35
17.3 Indicative Release Roadmap.....	35
18. Risks & Mitigations.....	36
19. Assumptions, Dependencies & Open Questions.....	37
19.1 Assumptions.....	37
19.2 Dependencies.....	37
19.3 Open Questions.....	37
20. Appendix — Glossary.....	38

1. Executive Summary

PanditConnect is a multi-sided online marketplace that connects devotees with verified Pandit Ji / Purohits for poojas, sanskars, yagnas and other religious ceremonies. Today this market runs almost entirely on word-of-mouth: discovery is slow, pricing is opaque, required pooja materials are unclear, and there is no credible way to judge a Pandit's quality before booking.

The platform solves three core problems — discovery, trust and pricing transparency — by letting customers search by pooja type, compare Pandits on verified ratings, experience, language and starting price, and book in a few guided steps. Pandits gain online visibility, a reputation that compounds over time, and a dashboard to manage inquiries, services and revenue.

Scope of the MVP. Verified Pandit onboarding, customer onboarding, pooja-based search with a transparent ranking algorithm, a structured booking workflow with explicit acceptance, a multi-dimensional customer review system, and role-based dashboards. The MVP deliberately defers payments, chat, video and AI matching to later phases.

Technology posture. The entire MVP runs on free-tier cloud services (Next.js on Vercel, MongoDB Atlas, Cloudinary, Resend, OpenStreetMap), which keeps burn at near-zero while we validate demand. Section 13.10 documents the practical limits of these tiers and the migration triggers.

What makes this document senior-grade. Beyond features, it specifies the data model, a defensible ranking and rating methodology that handles the cold-start problem, a comprehensive edge-case catalogue (Section 13), and a security/compliance posture appropriate to handling Indian identity data under the DPDP Act, 2023.

2. Product Vision

2.1 Vision Statement

Make finding and booking a trusted, verified Pandit as simple and reliable as booking a cab — while preserving the dignity, devotion and personal trust that religious ceremonies deserve.

2.2 Mission

- Help devotees find the right Pandit quickly, with confidence.
- Help Pandits grow their practice through online visibility and a compounding reputation.
- Build trust through verified identities, structured reviews and a transparent ranking system.
- Keep the booking journey short, predictable and stress-free.
- Be vernacular-first: fully usable in Hindi, Gujarati and English.

2.3 Product Principles

These principles resolve trade-offs when requirements conflict:

Principle	What it means in practice
Trust by default	Identity verification, verified-booking-only reviews, and visible credibility signals are non-negotiable, not optional polish.
Simplicity over features	We ship the shortest path to a completed booking. Anything that does not serve that path waits for a later phase.
Vernacular-first	Every customer-facing surface works in Hindi, Gujarati and English from day one; English is never assumed.
Fair, explainable ranking	Ranking factors are documented and defensible. New Pandits get a fair chance; established Pandits are not unfairly displaced.
Mobile-first	The primary device is a mid-range Android phone on a patchy network. Pages must be light and resilient.
Respect the occasion	Tone, copy and imagery stay respectful of religious sentiment across communities.

3. Problem Statement & Market Context

3.1 Current Problems

Problem	Impact on the user
Finding a trusted Pandit is hard	Customers rely on relatives and neighbours; choice is limited and quality is a gamble.
No standard pricing	Customers cannot compare or budget; some are overcharged, others under-served.
Materials are unclear	Customers do not know what samagri to arrange, causing last-minute scrambling and incomplete rituals.
No credibility system	There is no way to verify experience, sampraday or past behaviour before committing.
Word-of-mouth dependency	Good Pandits stay invisible beyond their local circle; demand and supply rarely meet efficiently.

3.2 Target Market

- **Initial geography:** Gujarat and adjacent Tier-1/Tier-2 cities, chosen to match the launch language set (Gujarati + Hindi + English) and allow tight, city-by-city supply build-up.
- **Demand side:** Hindu households planning poojas for life events (housewarming, weddings, naming, mundan) and recurring/occasion-based rituals (Satyanarayan Katha, festivals).
- **Supply side:** Independent Pandits and small purohit groups seeking bookings beyond their existing network.

3.3 Why Now

Smartphone penetration, UPI-driven comfort with digital transactions, and post-2020 normalisation of booking services online have made an at-home services marketplace viable in this category. The gap is a trusted, category-specific platform — horizontal marketplaces do not capture sampraday, specialization, materials or vernacular needs.

4. Solution Overview

PanditConnect is a digital platform on which a customer can search, compare and book a Pandit based on pooja type, reviews, price, experience and language. The same platform gives Pandits a profile, a booking pipeline and a reputation engine; an Admin oversees verification, content and disputes.

The end-to-end loop

**Search Pooja → Compare Pandits → Book → Pandit Accepts → Pooja Completed
→ Review → Better Ranking**

Every completed-and-reviewed pooja improves the data quality of the marketplace, which improves discovery, which drives more bookings — the core flywheel the product is designed around.

5. Goals & Success Metrics

5.1 Business Goals

- Achieve marketplace liquidity in the launch city: enough verified Pandits per pooja type that most searches return bookable results.
- Validate willingness to book (and later, to pay a lead/commission fee).
- Establish trust signals strong enough that customers prefer the platform over word-of-mouth.

5.2 Key Performance Indicators (KPIs)

Metric	Definition	MVP Target (first 6 months)
Verified Pandit supply	Pandits who completed verification, per launch city	≥ 75 per city
Search success rate	% of searches returning ≥ 3 bookable Pandits	≥ 85%
Booking conversion	Confirmed bookings ÷ searches	≥ 8%
Pandit response time	Median minutes from request to accept/decline	≤ 60 minutes
Booking completion rate	Completed ÷ confirmed bookings	≥ 80%
Review coverage	Reviews submitted ÷ completed bookings	≥ 60%
Repeat booking rate	Customers with ≥ 2 bookings in 6 months	≥ 20%

5.3 North Star Metric

Completed and reviewed poojas per month. It captures both sides of the marketplace working together: a real ceremony delivered, and trust data created for the next customer.

6. Target Users & Personas

6.1 Persona 1 — Customer (Devotee)

Attribute	Detail
Who	Household decision-maker (often 30–60) planning a pooja or sanskar.
Goals	Find a trustworthy Pandit fast; know the price and required materials up front; book a convenient date.
Frustrations	Hidden pricing, unreliable availability, not knowing what samagri to buy, no quality signal.
Devices	Mid-range Android phone; comfortable in Gujarati/Hindi; English varies.

6.2 Persona 2 — Pandit Ji (Service Provider)

Attribute	Detail
Who	Independent Pandit/purohit, varying experience and specialization (sampraday).
Goals	More bookings, online visibility, a reputation that grows, and clarity on earnings.
Frustrations	Dependence on local network; idle days; no way to showcase expertise or set transparent prices.
Devices	Smartphone; may have limited time/literacy for complex tooling — dashboard must be simple.

6.3 Persona 3 — Admin (Platform Operator)

Attribute	Detail
Who	Internal operations/trust-and-safety staff.
Goals	Verify Pandits, moderate content and reviews, resolve disputes, keep the catalogue clean.
Needs	Verification queue, review-moderation tools, user management, and audit visibility.

6.4 Supported Pooja & Sanskar Catalogue (launch set)

The MVP launches with a curated, normalised catalogue so that search, ranking and materials lists stay consistent across Pandits. Free-text pooja names are mapped to these canonical entries (see Section 13.3).

Pooja / Sanskar	Pooja / Sanskar	Pooja / Sanskar
Satyanarayan Katha	Griha Pravesh	Vivah Sanskar

Pooja / Sanskar	Pooja / Sanskar	Pooja / Sanskar
Rudrabhishek	Navchandi Yagna	Ganesh Pooja
Lakshmi Pooja	Vastu Pooja	Mundan Sanskar
Naming Ceremony (Namkaran)	—	—

7. System Architecture

7.1 Architecture Diagram

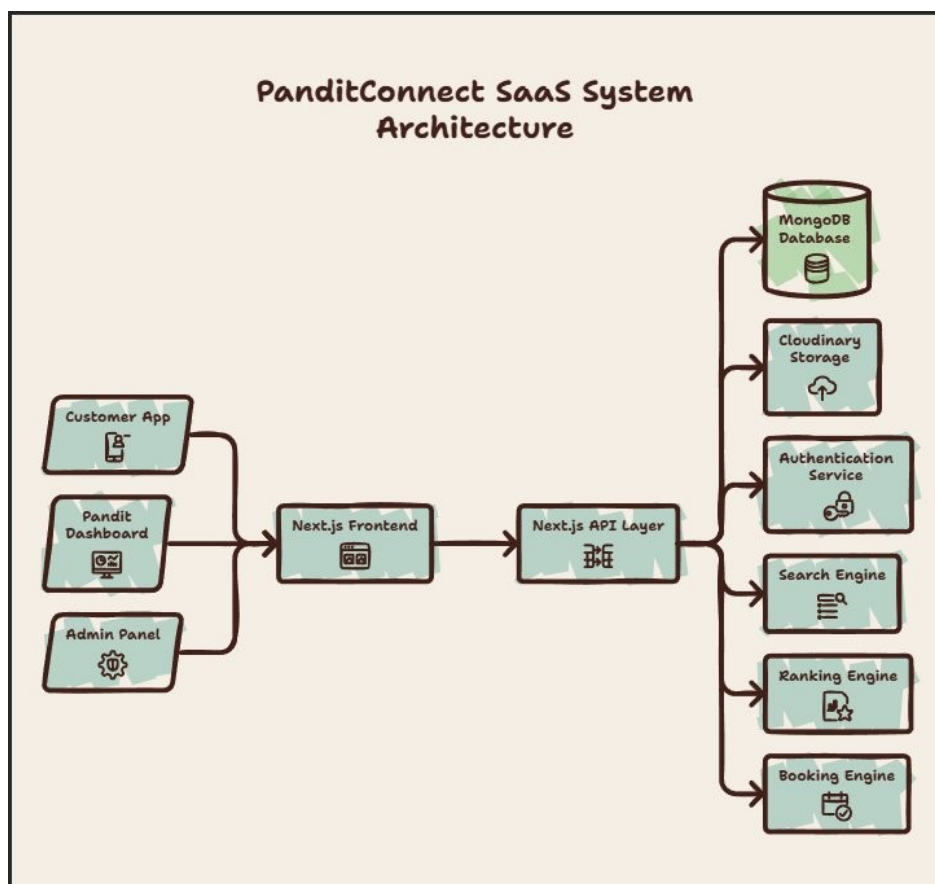


Figure 1 — PanditConnect high-level system architecture.

7.2 Component Responsibilities

Component	Responsibility
Client surfaces	Customer app, Pandit dashboard and Admin panel — all rendered by a single Next.js application with role-aware routing.
Next.js frontend	Server-rendered, mobile-first UI; localisation (i18next); calls API routes and Server Actions.
Next.js API layer	Server Actions + API routes for auth, search, booking, reviews, dashboard; enforces authorization on every call.
MongoDB (Atlas)	Primary datastore for users, pandits, poojas, materials, bookings, reviews, payments (Section 12).
Cloudinary	Stores and serves profile photos, verification images and (later) ceremony photos; handles transforms.
Authentication service	NextAuth.js sessions, role assignment, OTP/email verification.

Component	Responsibility
Search engine	Resolves pooja queries to the canonical catalogue and assembles candidate Pandits, filtered by service area and availability.
Ranking engine	Scores and orders candidates using the weighted, normalised formula in Section 9.5.
Booking engine	Owens the booking state machine, slot/conflict checks, acceptance timeouts and status transitions (Sections 9.7, 13.4).

7.3 Technology Stack & Rationale

Layer	Choice	Why this choice
Frontend	Next.js 15, TypeScript, Tailwind CSS, shadcn/ui	One codebase for SSR + API; type safety; fast, consistent UI primitives.
Backend	Next.js API Routes + Server Actions	Co-located with the frontend; no separate service to operate for the MVP.
Database	MongoDB Atlas (free tier)	Flexible schema suits evolving entities; generous free tier; easy Vercel integration.
Auth	NextAuth.js	Sessions, providers and role handling out of the box.
Media	Cloudinary (free tier)	Managed image storage, CDN delivery and on-the-fly transforms.
Maps / geo	OpenStreetMap	No API-key cost for service-area display and basic geocoding.
Email	Resend (free tier)	Transactional email for verification and booking notifications.
i18n	i18next	Mature localisation with namespacing and runtime language switching.
Hosting	Vercel (free tier)	Native Next.js deployment, preview builds, edge network.

Senior note — known constraints. Server Actions and serverless functions are stateless and time-limited; long jobs (image processing, bulk email) must be chunked or queued. Free-tier cold starts add latency; critical paths (search, booking) should stay lightweight. MongoDB free tier (shared, ~512 MB) is fine for validation but is a migration trigger as data grows — see 13.10.

8. User Roles & Access Control (RBAC)

Three roles exist: Customer, Pandit Ji and Admin. Authorization is enforced server-side on every API call and Server Action — the client never decides permissions. A single account holds exactly one role; a person who is both a customer and a Pandit uses separate accounts (see 13.1).

Capability	Customer	Pandit Ji	Admin
Register / manage own account	Yes	Yes	Yes
Search & view Pandit profiles	Yes	View only	Yes
Create / cancel a booking	Yes	—	On behalf (support)
Accept / decline / complete a booking	—	Yes (own)	Override (support)
Add / edit / delete pooja & materials	—	Yes (own)	Yes
Submit a review	Yes (own completed booking)	—	—
Respond to a review	—	Yes (reviews on self)	—
Verify Pandit identity	—	—	Yes
Moderate / remove reviews & content	—	Flag only	Yes
Manage users / roles	—	—	Yes
View revenue analytics	—	Own only	Platform-wide

9. Feature Specifications

Each feature lists its purpose, key requirements and acceptance criteria (AC). Edge-case behaviour is consolidated in Section 13 and cross-referenced.

9.1 Authentication & Onboarding

Purpose. Securely register and authenticate Customers, Pandits and Admins, and verify ownership of contact details.

Requirements

- Sign-up and login via mobile number with OTP and/or email + password, using NextAuth.js.
- Role is chosen at sign-up (Customer or Pandit); Admin accounts are provisioned internally.
- Mobile and email ownership are verified (OTP / verification link) before the account is fully active.
- Password reset via verified email/phone; sessions expire and can be revoked.

Acceptance Criteria

- AC1: A user cannot complete sign-up without verifying at least one contact channel.
- AC2: OTP expires after a fixed window and is rate-limited (see 13.1).
- AC3: Login state persists across reloads and is invalidated on logout/expiry.

9.2 Pandit Registration & Verification

Purpose. Onboard Pandits with enough structured data to power profiles, search and trust signals — and confirm identity before they become bookable.

Required fields

Group	Fields
Personal	Full name, photo, age, gender, mobile, email, address.
Religious	Sampraday, specialization, years of experience.
Languages	Any of Hindi, Gujarati, English.
Services	Per pooja: name (from catalogue), price, duration, description.
Materials	Per pooja: item name, quantity, notes.
Verification	Aadhaar verification, photo verification.

Verification workflow

1. Pandit submits profile + identity documents; status becomes Pending Verification.
2. Profile is not discoverable in search while pending (configurable preview).
3. Admin reviews identity and photo; approves or rejects with a reason.
4. On approval the profile becomes Verified and discoverable; on rejection the Pandit is notified and may resubmit.

Acceptance Criteria

- AC1: A Pandit cannot receive bookings until Verified.
- AC2: Identity numbers are never displayed back in full; only verification status is shown (see 14.2).
- AC3: Required fields are validated; incomplete profiles can be saved as Draft but not submitted.

9.3 Customer Registration

Purpose. Let customers create a lightweight account to book and review.

- **Required:** name, mobile, email, address, preferred language.
- **Optional:** family details, saved addresses, booking history (system-generated).
- **AC:** A customer can book immediately after verifying contact details; address can be supplied at booking time if omitted at sign-up.

9.4 Search & Discovery

Purpose. Turn a pooja need into a ranked list of bookable Pandits.

Behaviour

- Customer enters a pooja (e.g., “Satyanarayan Katha”); free text is resolved to a canonical catalogue entry (fuzzy + multilingual).
- Results are filtered to Pandits who offer that pooja, serve the customer's area, and are Verified.
- Optional filters: language, price range, date availability, minimum rating.
- Results are ordered by the ranking engine (9.5) and shown as profile cards.

Profile card contents

Photo, name, experience, languages, average rating (with review count), starting price, service areas, and actions: Call, Chat (Phase 2), Book Now, View Profile.

9.5 Ranking Algorithm

Purpose. Order search results fairly and transparently.

The base ranking score combines five signals with the following weights:

Signal	Weight	Notes
Average rating	40%	Bayesian-weighted (Section 10.5) so few reviews cannot dominate.
Completed bookings	25%	Normalised, with diminishing returns (log scale) to avoid runaway leaders.
Experience (years)	20%	Normalised and capped.
Response rate	10%	Share of requests answered within SLA.
Recent activity	5%	Recency of last completed booking / login.

Senior note — normalisation is mandatory. Raw values live on different scales (rating 0–5, bookings 0 → hundreds). Each signal must be normalised to 0–1 before applying weights, otherwise ‘completed bookings’ silently swamps ‘rating’. Cold start: a Verified Pandit with zero reviews receives the platform-mean rating as a prior (10.5) and a small new-on-platform boost for a limited window so they are not buried. Ties break on response rate, then recency. Weights are stored as configuration so they can be tuned without a deploy.

Acceptance Criteria

- AC1: Two Pandits with identical inputs produce identical scores (deterministic).
- AC2: A brand-new Verified Pandit appears in results (not last) for relevant searches.
- AC3: Changing a weight in config changes ordering without code changes.

9.6 Pandit Public Profile

Editable by the Pandit: bio, photo, services, pricing, languages. System-owned: rating summary, review list, completed-booking count, verification badge. The profile is the canonical place a customer evaluates trust before booking.

9.7 Booking System

Purpose. Move a customer from ‘interested’ to a confirmed, then completed, ceremony with explicit Pandit acceptance.

Booking flow

Select Pandit → Select Date & Time → Enter Address → Confirm → Pandit Accepts → Confirmed → Completed → Review

Booking state machine

State	Meaning	Valid transitions
Requested	Customer submitted; awaiting Pandit response	→ Accepted, Declined, Expired, Cancelled
Accepted / Confirmed	Pandit accepted; slot reserved	→ Completed, Cancelled (with reason)
Declined	Pandit declined	(terminal)
Expired	No response within SLA window	(terminal; customer prompted to rebook)
Cancelled	Cancelled by either party / admin	(terminal; reason + actor recorded)
Completed	Ceremony done; review unlocked	→ Reviewed

Acceptance Criteria

- AC1: A confirmed slot cannot be double-booked for the same Pandit (see 13.4).
- AC2: A request with no response within the SLA auto-Expires and notifies the customer.

- AC3: Only a Completed booking unlocks the review action.
- AC4: Every cancellation records who cancelled, when and why.

9.8 Pandit Dashboard

Six sections give the Pandit operational control:

Section	Shows / allows
Today's Inquiries	New leads, pending requests, accepted requests.
Revenue Tracking	Total earnings, monthly earnings, completed bookings (cancellations/refunds excluded — 13.6).
Pooja Management	Add / edit / delete pooja; update materials.
Reviews	Ratings, comments, review analytics; respond to reviews.
Public Profile	Edit bio, photo, services, pricing, languages.
Settings	Language, notifications, account security.

9.9 Notifications

- Transactional email via Resend for: contact verification, new booking request (to Pandit), accept/decline/expire (to customer), reminders, and review requests after completion.
- WhatsApp/SMS notifications are a Phase-2 enhancement.
- AC: Every state transition in the booking machine emits exactly one notification to the relevant party; duplicates are suppressed (idempotent sends).

9.10 Internationalisation (i18n)

- Hindi, Gujarati and English supported at launch via i18next; language is a user preference and switchable at runtime.
- UI strings are externalised; dates, numbers and currency follow the active locale.
- User-generated content (bio, reviews) is shown in the language it was written in, with a clear label; machine translation is out of scope for the MVP (13.9).
- AC: No hard-coded user-facing strings; a missing translation falls back to English rather than showing a key.

10. Customer Review & Rating System

The review system is the trust engine of the marketplace. It must produce honest, comparable, abuse-resistant signals that both rank Pandits fairly and help the next customer choose. This section specifies it end-to-end.

10.1 Objectives

- Capture structured, multi-dimensional feedback after every completed pooja.
- Guarantee authenticity: only customers who actually completed a booking can review (verified-booking-only).
- Aggregate ratings fairly so that a Pandit with few reviews is neither unfairly buried nor unfairly inflated.
- Give Pandits a right of reply and a clean dispute/flag path.
- Resist fake reviews, review-bombing and self-reviews.

10.2 Rating Dimensions

Each review captures four sub-dimensions plus an explicit overall score, all on a 1–5 star scale. The sub-dimensions help customers and surface why a Pandit is rated as they are.

Dimension	What it measures	Scale
Ritual Knowledge	Correctness and depth of the vidhi performed	1–5 ★
Punctuality	Arrived and finished on time	1–5 ★
Behaviour	Conduct, respectfulness, demeanour	1–5 ★
Communication	Clarity before, during and after the pooja	1–5 ★
Overall Satisfaction	Headline score (drives ranking)	1–5 ★

- **Optional written comment** (length-bounded; sanitised — see 10.7).
- **Optional photo of the ceremony** (moderated; Phase-1.5).
- **Required:** the Overall score. Sub-dimensions may be optional to reduce friction; if omitted, only Overall feeds ranking.

10.3 Eligibility & Rules

Rule	Behaviour
Verified booking only	A review can be created only by the customer of a booking in Completed state.
One review per booking	Exactly one review per (customer, booking). A second attempt edits the existing review.
Edit window	The author may edit the review for 7 days after submission; after that it locks (audit retained).
No self-review	A Pandit account cannot review itself, and a booking where customer and Pandit resolve to the same person is rejected (see 13.5).

Rule	Behaviour
Review window	Reviews can be submitted up to 30 days after completion; later attempts are blocked and the customer is informed.
Account deletion	If a reviewer deletes their account, reviews are anonymised (“Verified customer”) rather than deleted, to preserve rating integrity (subject to 14).

10.4 Review Lifecycle

Booking Completed → Review Prompt → Submit (Pending Moderation) → Published → Pandit Reply (optional)

Reviews may pass an automated moderation check on submission (profanity / spam / PII filters). Clean reviews publish immediately; flagged reviews enter an Admin queue. Any party can subsequently flag a published review, which re-opens moderation.

10.5 Rating Aggregation & Anti-Gaming

A simple arithmetic mean is unfair at low review counts: one 5★ review would outrank a Pandit with 200 reviews averaging 4.7★. PanditConnect therefore uses a Bayesian (shrinkage) average for the ranking score:

$WR = (v / (v + m)) \times R + (m / (v + m)) \times C$	
Symbol	Meaning
WR	Weighted rating used by the ranking engine
R	This Pandit's mean Overall rating
v	Number of reviews for this Pandit
m	Minimum-reviews threshold / prior strength (tunable, e.g. 5)
C	Platform-wide mean Overall rating (e.g. 4.2)

Worked example. Pandit A: $R = 5.0$ with $v = 1$. Pandit B: $R = 4.7$ with $v = 200$. With $m = 5$ and $C = 4.2$: $WR(A) = (1/6) \times 5.0 + (5/6) \times 4.2 = 4.33$; $WR(B) = (200/205) \times 4.7 + (5/205) \times 4.2 = 4.69$. Pandit B correctly ranks higher despite a lower raw star count, while a genuinely excellent newcomer still climbs quickly as reviews accumulate.

- **Display vs ranking:** the profile shows the honest raw average and the review count; the Bayesian WR is used only for ordering, so we never mislead the customer.
- **Time decay (optional):** recent reviews may be weighted slightly higher so ratings reflect current quality.
- **Recompute:** WR and C are recomputed on review create/edit/remove and cached on the Pandit record.

10.6 Pandit Response

- A Pandit may post exactly one public reply per review (editable within its own window).

- Replies are moderated by the same filters and shown beneath the review.
- A reply does not change the score; it lets the Pandit add context or address concerns.

10.7 Moderation & Abuse Prevention

Threat	Control
Fake / paid reviews	Verified-booking-only gating; one review per completed booking.
Self-review (sock puppet)	Block customer==Pandit identity; flag shared device/contact/payment fingerprints for Admin review (13.5).
Review bombing	Rate-limit reviews per account; anomaly detection on rating velocity; Bayesian averaging dampens short bursts.
Offensive / PII content	Profanity + PII filters on submit; reject or queue; never publish raw phone/Aadhaar in text.
Retaliatory edits	Edit window + immutable audit trail of original and edits.
Disputed reviews	Pandit can flag; Admin can hide, annotate or remove with reason; outcomes logged.
XSS via review text	Treat all review/reply text as untrusted; sanitise and escape on render (14.3).

10.8 Display & UX

- Profile shows: overall average (raw), total review count, a 1–5 star distribution bar, and per-dimension averages.
- Review list is sortable (most recent / most helpful) and shows a “Verified booking” badge, dimension chips, comment, optional photo, date and any Pandit reply.
- Empty state for new Pandits: “New on PanditConnect — be the first to review” rather than “0.0 ★”, which reads as poor quality (13.5).
- AC: A customer can never submit a review for a booking that is not theirs or not Completed; the review CTA is otherwise hidden.

10.9 Sample Reviews (illustrative)

The following examples illustrate the intended data shape and tone shown on a Pandit profile.

Reviewer	Pooja	Ratings	Comment & Pandit reply
R. Sharma <i>Verified booking</i>	Griha Pravesh	Overall 5★ Knowledge 5 · Punctuality 5 · Behaviour 5 · Comm. 4	Explained every step in Gujarati and arrived 10 minutes early. Sent the samagri list two days before — nothing was missing. <i>Reply: Thank you, it was a pleasure to serve your family. — Pandit ji</i>
A. Patel <i>Verified booking</i>	Satyanarayan Katha	Overall 4★ Knowledge 5 · Punctuality 3 Behaviour 4 ·	Very thorough katha. Reached about 20 minutes late due to traffic but informed us in advance, so we could adjust.

Reviewer	Pooja	Ratings	Comment & Pandit reply
		Comm. 4	
Verified customer <i>name hidden</i>	Mundan Sanskar	Overall 5★ Knowledge 5 · Punctuality 5 Behaviour 5 · Comm. 5	Gentle and patient with the baby; guided us through the whole vidhi calmly. Highly recommend.

11. User Journey Flows

11.1 Customer Journey



Figure 2 — End-to-end customer journey, from registration to review.

11.2 Customer Flow (detailed)

Register/Login → Search Pooja → View Pandits → Compare Ratings → View Profile → Book → Confirmation → Pooja Completed → Review

11.3 Pandit Onboarding Flow

Register → Add Profile & Services → Submit Identity → Admin Verifies → Profile Goes Live → Receive Bookings

Cross-references: booking-flow edge cases in 13.4; verification edge cases in 13.2.

12. Data Model

12.1 Entities & Relationships

- A users document holds identity and role for everyone (Customer, Pandit, Admin).
- A pandits document extends a user (1:1) with professional details and cached rating fields.
- A pandit offers many poojas (1:N); each pooja has many materials (1:N).
- A booking links a customer, a pandit and a pooja; a completed booking yields at most one review.
- A payment (Phase 2) links to a booking (1:1).

The schemas below specify each collection. Types are indicative; all documents carry createdAt / updatedAt and soft-delete flags where relevant (13.7).

12.2 Collection Schemas

users

Field	Type	Req	Description
_id	ObjectId	Yes	Primary key
role	enum	Yes	customer pandit admin
name	string	Yes	Display name
email	string	Yes*	Unique; verified flag stored separately
phone	string	Yes*	Unique; E.164; verified flag separate
preferredLanguage	enum	Yes	hi gu en
status	enum	Yes	active pending suspended deleted

pandits

Field	Type	Req	Description
userId	ObjectId	Yes	References users._id (unique)
experienceYears	number	Yes	Years of practice
sampraday	string	Yes	Tradition/lineage
specialization	string[]	Yes	Pooja specialities
languages	enum[]	Yes	Subset of {hi, gu, en}
serviceAreas	string[]/geo	Yes	Areas/pincodes served
profilePhoto	string(url)	Yes	Cloudinary URL
verificationStatus	enum	Yes	pending verified rejected
ratingAvg	number	No	Cached raw mean Overall
ratingCount	number	No	Cached review count
ratingWeighted	number	No	Cached Bayesian WR (10.5)

Field	Type	Req	Description
responseRate	number	No	% requests answered within SLA

poojas

Field	Type	Req	Description
_id	ObjectId	Yes	Primary key
panditId	ObjectId	Yes	Owning Pandit
catalogKey	string	Yes	Canonical catalogue id (Section 6.4)
name	string	Yes	Display name
description	string	No	Details of the service
price	number	Yes	Starting price (INR)
durationMin	number	Yes	Typical duration in minutes
active	boolean	Yes	Listed / unlisted

materials

Field	Type	Req	Description
_id	ObjectId	Yes	Primary key
poojaId	ObjectId	Yes	Parent pooja
itemName	string	Yes	Samagri item
quantity	string	Yes	Amount/unit
notes	string	No	Optional guidance

bookings

Field	Type	Req	Description
_id	ObjectId	Yes	Primary key
customerId	ObjectId	Yes	Booking customer
panditId	ObjectId	Yes	Booked Pandit
poojaId	ObjectId	Yes	Selected pooja
scheduledAt	datetime	Yes	Date + time (with timezone)
address	object	Yes	Service location
status	enum	Yes	requested confirmed declined expired cancelled completed
cancellation	object	No	{ by, reason, at } when cancelled
respondedAt	datetime	No	When Pandit accepted/declined

reviews

Field	Type	Req	Description
_id	ObjectId	Yes	Primary key
bookingId	ObjectId	Yes	Unique — one review per booking
customerId	ObjectId	Yes	Author
panditId	ObjectId	Yes	Subject
overall	number	Yes	1–5 (drives ranking)
ritualKnowledge / punctuality / behaviour / communication	number	No	1–5 each
comment	string	No	Sanitised free text
photoUrl	string	No	Moderated ceremony photo
status	enum	Yes	pending published hidden removed
panditReply	object	No	{ text, at } single reply

payments (Phase 2)

Field	Type	Req	Description
_id	ObjectId	Yes	Primary key
bookingId	ObjectId	Yes	Related booking (unique)
amount	number	Yes	Amount (INR)
status	enum	Yes	pending paid refunded failed

12.3 Indexing & Integrity Strategy

- Unique indexes: users.email, users.phone, pandits.userId, reviews.bookingId.
- Query indexes: poojas.catalogKey, pandits.serviceAreas (geo), bookings.{panditId, scheduledAt} for conflict checks, reviews.panditId.
- Referential integrity is enforced in the application layer (MongoDB has no FKs); deletes cascade via soft-delete + cleanup jobs (13.7).
- Rating fields on pandits are denormalised caches, recomputed transactionally on review changes.

13. Edge Cases & Error Handling

This catalogue is the heart of a build-ready specification. Each row pairs a scenario with the expected system behaviour. QA should treat these as test cases; engineering should treat them as requirements, not afterthoughts.

13.1 Authentication & Accounts

Scenario	Expected behaviour
Email/phone already registered	Block sign-up with a clear message; offer login / password reset. Never reveal which field matched in a way that aids enumeration.
OTP expired or wrong	Reject; allow resend after cooldown; cap attempts, then temporary lockout to stop brute force.
OTP request spamming	Rate-limit OTP requests per number/IP; exponential backoff.
Same person is Customer and Pandit	Supported via two separate accounts on different identifiers; a single account holds one role only.
Concurrent logins / stale session	Sessions expire; logout/expiry invalidates tokens; sensitive actions re-check the session server-side.
Forgot password	Reset only via a verified channel; reset links are single-use and time-bound.
Account suspended/deleted tries to act	All authenticated calls re-check status and refuse with an explanation.

13.2 Registration & Verification

Scenario	Expected behaviour
Incomplete Pandit profile	Saveable as Draft; cannot be Submitted until required fields pass validation.
Invalid / mismatched identity docs	Admin rejects with a reason; Pandit notified and may resubmit; status stays not-discoverable.
Pandit stuck in Pending	Verification SLA + reminders to Admin; Pandit sees clear pending state, not silence.
Duplicate Pandit (same person, new account)	Identity/contact fingerprints flagged for Admin to merge or reject.
Age below threshold / invalid phone format	Field-level validation blocks submission with specific guidance.
Photo fails moderation	Upload rejected; Pandit asked to supply a compliant photo.

13.3 Search & Ranking

Scenario	Expected behaviour
No Pandits for the pooja	Show an empty state with suggestions (nearby areas, related poojas, 'notify me') instead of a blank screen.
No Pandits in the customer's area	Offer to widen the radius or show closest available areas; never silently return zero.
All candidates unavailable on chosen date	Show available alternative dates / Pandits; keep the pooja filter intact.
Misspelled or vernacular pooja name	Fuzzy + multilingual matching maps free text to the canonical catalogue; show 'did you mean...'.
Equal ranking scores	Deterministic tie-break: response rate, then recency, then stable id ordering.
Brand-new Pandit (0 reviews)	Uses platform-mean prior + limited new-on-platform boost so they appear, not last (10.5).
Signal missing (e.g., no responseRate yet)	Treat as neutral/default in normalisation; never crash or drop the candidate.

13.4 Booking

Scenario	Expected behaviour
Double-booking the same slot	Atomic conflict check on {panditId, scheduledAt}; the second request is rejected with alternatives.
Booking a past date/time	Blocked at validation; only future slots (with a minimum lead time) are selectable.
Booking too far in advance	Cap the horizon (configurable) to keep schedules realistic.
Pandit never responds	Auto-Expire at SLA; notify the customer and prompt to rebook with another Pandit.
Pandit accepts, then becomes unavailable	Pandit can Cancel-with-reason; customer is notified, slot is freed, rebooking suggested; cancellation counts against reliability metrics.
Customer cancels last-minute	Allowed with reason; recorded; (Phase 2) cancellation policy/fees may apply.
Customer books several Pandits for one slot	Permitted as separate requests, but the customer is warned; confirming one does not auto-cancel others (explicit action required).
Address outside service area	Warn before confirm; block or require Pandit opt-in for out-of-area service.
Festival peak / capacity	Availability reflects the Pandit's real calendar; fully-booked slots are not offered.
Time-zone / DST ambiguity	Store UTC + the booking's local timezone; display in the user's locale to avoid off-by-one errors.

Scenario	Expected behaviour
Payment fails mid-booking (Phase 2)	Booking is not confirmed; slot is released; customer can retry without creating duplicates (idempotency keys).

13.5 Reviews

Scenario	Expected behaviour
Review before completion	Blocked; the review CTA is unavailable until status = Completed.
Non-customer tries to review	Rejected server-side; only the booking's customer may review it.
Second review for same booking	Edits the existing review (one per booking), within the edit window.
Self-review / sock puppet	customer==Pandit rejected; shared-fingerprint reviews flagged for Admin (10.7).
Review bombing / velocity spike	Per-account rate limits + anomaly detection; Bayesian average dampens bursts (10.5).
Offensive language / PII in text	Filtered on submit; rejected or queued; never published with raw PII.
Reviewer deletes account	Review anonymised ("Verified customer"), not deleted, to protect rating integrity.
New Pandit, zero reviews	Show 'New on PanditConnect' empty state, not '0.0★'.
Disputed review	Pandit flags → Admin reviews → hide/annotate/remove with reason, all logged.
One 5★ vs many 4.7★	Bayesian weighting ranks the high-volume Pandit correctly while display stays honest (10.5).

13.6 Dashboard & Revenue

Scenario	Expected behaviour
Cancelled/refunded booking in earnings	Excluded from earnings and completed counts; only Completed (and Phase-2 paid) bookings count.
Deleting a pooja with active bookings	Block hard delete; allow unlist so existing bookings remain valid and historical.
'Today' boundary across timezones	'Today' is computed in the Pandit's locale, consistently.
Concurrent edits to the same listing	Last-write-wins with updatedAt guard, or optimistic-concurrency rejection on stale writes (13.7).

13.7 Data Integrity & Concurrency

Scenario	Expected behaviour
Deleting a Pandit	Soft-delete; cascade soft-delete/anonymise poojas, materials and reviews; bookings retained for history.
Orphaned references	Cleanup jobs and application-layer integrity checks; reads tolerate missing optional refs.
Simultaneous writes to one document	Optimistic concurrency via updatedAt/version; stale writes rejected and retried.
Stale denormalised ratings	Rating caches recomputed on every review change; periodic reconciliation job as a safety net.

13.8 Media & Uploads

Scenario	Expected behaviour
Oversized or wrong-format image	Validate type/size client- and server-side; reject with guidance; compress on upload.
Upload fails mid-way	Surface a retry; never leave a half-saved record referencing a missing asset.
Inappropriate image	Moderation gate before publishing profile/ceremony photos.
Cloudinary quota reached	Graceful fallback (queue/placeholder) + alert; see 13.10.

13.9 Internationalisation

Scenario	Expected behaviour
Missing translation string	Fall back to English; never render a raw key to the user.
Mixed-language user content	Show content in its original language with a label; no auto-translation in the MVP.
Locale-specific dates/numbers/currency	Format per active locale; store canonical values internally (UTC, INR minor units).

13.10 Infrastructure & Free-Tier Limits

Scenario	Expected behaviour / trigger
MongoDB Atlas free tier (~512 MB, shared)	Monitor size/throughput; migration trigger to a paid M-tier before limits bite; avoid unbounded growth (e.g., logs in app DB).
Vercel free-tier limits / cold starts	Keep critical paths light; watch function duration and bandwidth; plan upgrade when traffic warrants.
Resend free-tier email cap	Throttle/queue non-critical mail; prioritise transactional sends; alert near the cap.
Cloudinary free-tier cap	Optimise/transform on upload; alert near limits; plan upgrade for media growth.

Scenario	Expected behaviour / trigger
Third-party outage (email/maps/media)	Degrade gracefully: queue emails, fall back maps to text addresses, show placeholders for media.

14. Security, Privacy & Compliance

14.1 Data Protection

- All traffic over HTTPS/TLS; secrets in environment variables, never in the client bundle or repo.
- Sensitive data encrypted in transit and at rest; least-privilege access to the database.
- Audit trails for verification decisions, review moderation and admin overrides.

14.2 Identity (Aadhaar) & PII Handling

- Treat Aadhaar and similar identifiers as highly sensitive. Store verification status and, at most, masked/last-digits — not full numbers in the clear.
- Never display identity numbers back to any user, including the Pandit; show only 'Verified'.
- Phone numbers are private by default; the 'Call' action should use masking/proxy where feasible rather than exposing raw numbers.
- **Compliance:** align with India's Digital Personal Data Protection Act, 2023 — purpose limitation, consent for processing, data-retention limits, and a deletion path (which anonymises reviews, per 10.3).

14.3 Application Security

- Authorise every API route and Server Action server-side by role and ownership (Section 8); never trust client claims.
- Validate and sanitise all input; escape all user-generated content on render to prevent XSS (reviews, bios, replies).
- Guard against NoSQL injection (typed queries, no unsanitised object spreads into queries).
- Rate-limit auth, OTP, search and review endpoints; add bot defences on public forms.
- Idempotency keys on booking/payment operations to prevent duplicates on retries.

14.4 Content & Trust Safety

Image and text moderation pipelines (Section 10.7, 13.8) protect both customers and Pandits; an Admin audit log records moderation actions and reasons.

15. Non-Functional Requirements

Area	Requirement
Performance	Search and profile pages target a fast first contentful paint on mid-range Android over 4G; critical API calls respond within a couple of hundred milliseconds at the p95 under MVP load.
Availability	Best-effort high availability on managed free tiers; graceful degradation when a dependency is down (13.10).
Scalability	Stateless app layer scales horizontally on Vercel; database is the first scaling constraint and has a defined migration trigger.
Reliability	Idempotent notifications and booking operations; no duplicate side-effects on retry.
Accessibility	Sufficient colour contrast, keyboard navigability, and legible type sizes; vernacular fonts render correctly.
Compatibility	Modern mobile and desktop browsers; mobile-first layouts.
Observability	Application logging and basic error monitoring; key funnels (search → book, book → complete → review) instrumented for the KPIs in Section 5.
Maintainability	TypeScript throughout; configuration (ranking weights, thresholds, caps) externalised from code.

16. MVP Scope

16.1 Must Have

Capability	Included
Authentication (OTP/email)	Yes
Pandit registration + verification	Yes
Customer registration	Yes
Pooja-based search	Yes
Transparent ranking system	Yes
Booking system with acceptance flow	Yes
Multi-dimensional review system	Yes
Pandit dashboard	Yes
Hindi / Gujarati / English i18n	Yes

16.2 Nice to Have (later)

- In-app chat between customer and Pandit.
- Video consultation.
- Online payments and refunds.
- AI-based Pandit matching / recommendations.
- WhatsApp/SMS notifications.

16.3 Explicitly Out of Scope for the MVP

- Live-streaming of poojas, digital prasad tracking, astrology/kundli services.
- Marketplace for samagri/materials purchase.
- Native mobile apps (the MVP is a responsive web app).

Guiding mantra: *Build the MVP first. Validate the market. Scale later.*

17. Monetisation & Roadmap

17.1 Revenue Model (phased)

Phase	Model	Rationale
Phase 1	Free platform	Maximise supply and demand; build trust data and liquidity.
Phase 2	Lead commission	Charge per qualified lead/booking once value is proven.
Phase 3	Premium Pandit profile	Subscription for enhanced profile features and insights.
Phase 4	Featured listings	Paid placement, clearly labelled, without distorting honest ranking.

17.2 Phase-2 Feature Candidates

- Video consultation, online pooja booking and live streaming.
- Digital prasad tracking; astrology and kundli services.
- WhatsApp notifications; AI recommendation engine.

17.3 Indicative Release Roadmap

Milestone	Focus
M1 — Foundations	Auth, roles, customer + Pandit registration, verification queue.
M2 — Discovery	Catalogue, search, ranking engine, public profiles.
M3 — Transactions	Booking state machine, notifications, Pandit dashboard.
M4 — Trust	Review system + Bayesian ranking, moderation, dispute handling.
M5 — Hardening	Edge-case coverage (Section 13), security pass, i18n completeness, KPI instrumentation.

18. Risks & Mitigations

Risk	Likelihood	Impact	Mitigation
Cold-start: too few Pandits per pooja/area	High	High	City-by-city manual supply onboarding before opening demand; 'notify me' to gauge latent demand.
Trust failure (a bad ceremony experience)	Medium	High	Strict verification, verified-only reviews, dispute handling, reliability metrics in ranking.
Fake reviews / gaming	Medium	High	Verified-booking gating, Bayesian averaging, anomaly detection (Section 10.7).
Free-tier limits hit during growth	Medium	Medium	Monitoring + defined migration triggers (13.10).
PII / compliance breach	Low	High	DPDP-aligned handling, masking, encryption, least privilege (Section 14).
Pandit non-responsiveness	Medium	Medium	Response-time SLA + auto-expire + response-rate weighting in ranking.
Scope creep delaying MVP	Medium	Medium	Strict Must/Nice/Out-of-scope discipline (Section 16).

19. Assumptions, Dependencies & Open Questions

19.1 Assumptions

- Customers are willing to book at-home religious services through an app once trust signals are credible.
- Enough Pandits in the launch city will onboard and keep availability current.
- Free-tier infrastructure is sufficient to validate demand before monetisation.
- The curated catalogue (Section 6.4) covers the large majority of launch-market demand.

19.2 Dependencies

- Third-party services: Vercel, MongoDB Atlas, Cloudinary, Resend, OpenStreetMap, NextAuth.js.
- An identity-verification method/provider for Aadhaar/photo checks.
- Operational capacity for Admin verification and moderation at launch volumes.

19.3 Open Questions

- Exact identity-verification provider and the precise data we are permitted to store.
- Cancellation policy and any fees (relevant once payments arrive in Phase 2).
- Final values for ranking weights and the Bayesian m / C parameters after early data.
- Whether sub-dimension ratings are required or optional at launch (friction vs. signal).
- Service-area model: pincode list vs. geo-radius for matching.

20. Appendix — Glossary

Domain terms used throughout this document, for engineers and stakeholders less familiar with the subject matter.

Term	Meaning
Pandit Ji / Purohit	A priest who performs Hindu rituals and ceremonies.
Pooja	A worship ritual offered to a deity.
Sanskar	A rite of passage (e.g., naming, mundan, marriage).
Sampraday	A religious tradition or lineage a Pandit belongs to.
Samagri	The materials/items required to perform a pooja.
Vidhi	The prescribed procedure/method of a ritual.
Satyanarayan Katha	A popular pooja with the recitation of the Satyanarayan story.
Griha Pravesh	A housewarming ceremony for entering a new home.
Rudrabhishek	An abhishek ritual dedicated to Lord Shiva.
Navchandi Yagna	A fire ritual invoking the Goddess (Chandi).
Mundan Sanskar	A child's first head-shaving ceremony.
Bayesian (shrinkage) average	A rating method that pulls low-count averages toward the platform mean for fairness (Section 10.5).
SLA	Service-level agreement — here, the time within which a Pandit should respond to a request.
RBAC	Role-Based Access Control — permissions assigned by role (Section 8).
DPDP Act, 2023	India's Digital Personal Data Protection Act governing personal-data processing.

— End of Document —